6.7 THE ĆUK CONVERTER

The Ćuk switching topology is shown in Fig. 6-13*a.* Output voltage magnitude can be either larger or smaller than that of the input, and there is a polarity reversal on the output.

The inductor on the input acts as a filter for the dc supply to prevent large harmonic content. Unlike the previous converter topologies where energy transfer is associated with the inductor, energy transfer for the Ćuk converter depends on the capacitor C_1.

The analysis begins with these assumptions:

1. Both inductors are very large and the currents in them are constant.
2. Both capacitors are very large and the voltages across them are constant.
3. The circuit is operating in steady state, meaning that voltage and current waveforms are periodic.
4. For a duty ratio of D, the switch is closed for time DT and open for $(1-D)T$.
5. The switch and the diode are ideal.

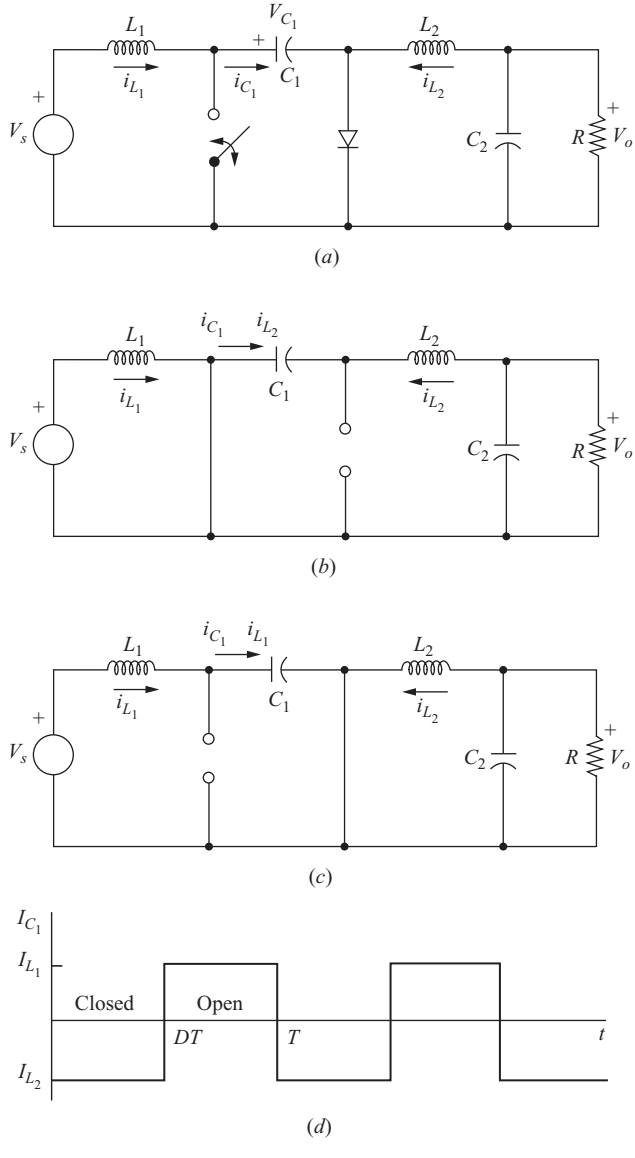


Figure 6-13 The Ćuk converter. (*a*) Circuit; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open; (*d*) Current in L_1 for a large inductance.

The average voltage across C_1 is computed from Kirchhoff's voltage law around the outermost loop. The average voltage across the inductors is zero for steady-state operation, resulting in

$$V_{C_1} = V_s - V_o$$

With the switch closed, the diode is off and the current in capacitor C_1 is

$$(i_{C_1})_{\text{closed}} = -I_{L_2} \tag{6-56}$$

With the switch open, the currents in L_1 and L_2 force the diode on. The current in capacitor C_1 is

$$(i_{C_1})_{\text{open}} = I_{L_1} \tag{6-57}$$

The power absorbed by the load is equal to the power supplied by the source:

$$-V_o I_{L_2} = V_s I_{L_1} \tag{6-58}$$

For periodic operation, the average capacitor current is zero. With the switch on for time DT and off for $(1-D)T$,

$$[(i_{C_1})_{\text{closed}}]DT + [(i_{C_1})_{\text{open}}](1-D)T = 0$$

Substituting using Eqs. (6-56) and (6-57),

$$-I_{L_2}DT + I_{L_1}(1-D)T = 0$$

or

$$\frac{I_{L_1}}{I_{L_2}} = \frac{D}{1-D} \tag{6-59}$$

Next, the average power supplied by the source must be the same as the average power absorbed by the load,

$$P_s = P_o$$

$$V_s I_{L_1} = -V_o I_{L_2} \tag{6-60}$$

$$\frac{I_{L_1}}{I_{L_2}} = -\frac{V_o}{V_s}$$

Combining Eqs. (6-59) and (6-60), the relationship between the output and input voltages is

$$\boxed{V_o = -V_s\left(\frac{D}{1-D}\right)} \tag{6-61}$$

The negative sign indicates a polarity reversal between output and input.

Note that the components on the output (L_2, C_2, and R) are in the same configuration as the buck converter and that the inductor current has the same form as for the buck converter. Therefore, the ripple, or variation in output voltage, is the same as for the buck converter:

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{1-D}{8L_2C_2f^2}} \tag{6-62}$$

The output ripple voltage will be affected by the equivalence series resistance of the capacitor as it was in the convertors discussed previously.

The ripple in C_1 can be estimated by computing the change in v_{C1} in the interval when the switch is open and the currents i_{L_1} and i_{C_1} are the same. Assuming the current in L_1 to be constant at a level I_{L_1} and using Eqs. (6-60) and (6-61), we have

$$\Delta v_{C_1} \approx \frac{1}{C_1}\int_{DT}^{T} I_{L_1} d(t) = \frac{I_{L1}}{C_1}(1-D)T = \frac{V_s}{RC_1 f}\left(\frac{D^2}{1-D}\right)$$

or

$$\boxed{\Delta v_{C_1} \approx \frac{V_o D}{RC_1 f}} \tag{6-63}$$

The fluctuations in inductor currents can be computed by examining the inductor voltages while the switch is closed. The voltage across L_1 with the switch closed is

$$v_{L_1} = V_s = L_1 \frac{di_{L_1}}{dt} \tag{6-64}$$

In the time interval DT when the switch is closed, the change in inductor current is

$$\frac{\Delta i_{L1}}{DT} = \frac{V_s}{L_1}$$

or

$$\boxed{\Delta i_{L1} = \frac{V_s DT}{L_1} = \frac{V_s D}{L_1 f}} \tag{6-65}$$

For inductor L_2, the voltage across it when the switch is closed is

$$v_{L_2} = V_o + (V_s - V_o) = V_s = L_2 \frac{di_{L_2}}{dt} \tag{6-66}$$

The change in i_{L_2} is then

$$\boxed{\Delta i_{L_2} = \frac{V_s DT}{L_2} = \frac{V_s D}{L_2 f}} \tag{6-67}$$

For continuous current in the inductors, the average current must be greater than one-half the change in current. Minimum inductor sizes for continuous current are

$$\boxed{\begin{aligned} L_{1,\min} &= \frac{(1-D)^2 R}{2Df} \\ L_{2,\min} &= \frac{(1-D)R}{2f} \end{aligned}} \tag{6-68}$$

EXAMPLE 6-7

Ćuk Converter Design

A Ćuk converter has an input of 12 V and is to have an output of −18 V supplying a 40-W load. Select the duty ratio, the switching frequency, the inductor sizes such that the change in inductor currents is no more than 10 percent of the average inductor current, the output ripple voltage is no more than 1 percent, and the ripple voltage across C_1 is no more than 5 percent.

■ Solution

The duty ratio is obtained from Eq. (6-61),

$$\frac{V_o}{V_s} = -\frac{D}{1-D} = \frac{-18}{12} = -1.5$$

or

$$D = 0.6$$

Next, the switching frequency needs to be selected. Higher switching frequencies result in smaller current variations in the inductors. Let $f = 50$ kHz. The average inductor currents are determined from the power and voltage specifications.

$$I_{L_2} = \frac{P_o}{-V_o} = \frac{40\,\text{W}}{18\,\text{V}} = 2.22\ \text{A}$$

$$I_{L_1} = \frac{P_s}{V_s} = \frac{40\,\text{W}}{12\,\text{V}} = 3.33\ \text{A}$$

The change in inductor currents is computed from Eqs. (6-65) and (6-67).

$$\Delta i_L = \frac{V_s D}{Lf}$$

The 10 percent limit in changes in inductor currents requires

$$L_2 \geq \frac{V_s D}{f \Delta i_{L_2}} = \frac{(12)(0.6)}{(50{,}000)(0.222)} = 649\ \mu\text{H}$$

$$L_1 \geq \frac{V_s D}{f \Delta i_{L_1}} = \frac{(12)(0.6)}{(50{,}000)(0.333)} = 432\ \mu\text{H}$$

From Eq. (6-62), the output ripple specification requires

$$C_2 \geq \frac{1-D}{(\Delta V_o / V_o) 8 L_2 f^2} = \frac{1-0.6}{(0.01)(8)(649)(10)^{-6}(50{,}000)^2} = 3.08\ \mu\text{F}$$

Average voltage across C_1 is $V_s - V_o = 12 - (-18) = 30$ V, so the maximum change in v_{C_1} is (30)(0.05) = 1.5 V.

The equivalent load resistance is

$$R = \frac{V_o^2}{P} = \frac{(18)^2}{40} = 8.1\,\Omega$$

Now C_1 is computed from the ripple specification and Eq. (6-63).

$$C_1 \geq \frac{V_o D}{R f \Delta v_{C_1}} = \frac{(18)(0.6)}{(8.1)(50{,}000)(1.5)} = 17.8\ \mu\text{F}$$